

Refill the Kettle and Dispenser at the Sink

SOP (1x Episode: kettle and dispenser tank, in situ)

One episode fills **both** things at the sink where they live. The kettle first. Then the dispenser. The kettle goes to the sink itself. The dispenser never moves, so a **separate fill jug** carries its water from the sink and pours it into the dispenser tank. The episode ends when the kettle and the tank both read full to their lines, the jug and the cloth are back on their spots, and the kettle and the jug are wiped dry. Never split the kettle and the dispenser into two episodes.

The base is **passive**. It has no motor. You push it by hand to the counter in front of the sink and lock it there. Nothing is carried away to a table. Everything the episode touches is already there when recording starts: the sink and its tap, the kettle on its power base at the back of the counter, the dispenser next to it, the fill jug on its spot at the right end of the counter, and the cloth beside the jug.

The sink is used **as found**. The tap is off and set to cold. The kettle stands on its power base, below its line. The dispenser tank is below its line. The fill jug stands empty on its spot. The counter is dry. The episode ends with the kettle and the tank full to their lines, the jug back on its spot, the tap off, and the kettle and the jug wiped dry.

This is an in situ task. Four rules come from that. First, there is a **wall cabinet right above the back of the counter**, so nothing is ever lifted high. You lift a thing just off what it stands on, bring it forward into the **front lane**, and carry it along the lane low and level. The one lift that goes higher is the pour, and even then the jug goes up only as far as its lip needs to clear the filler rim. Second, **the tap never runs into anything held in the air**. You stand the kettle or the jug on the bowl floor first, then turn the water on. You lift it only after the water is off. Third, **the fixtures never move**. You never lean on, push, drag, or lift the sink, the tap, the power base, or the dispenser. The dispenser tank never comes off its body. Fourth, **you never leave the water running**. The tap is on only while something stands under it, and you shut it off before you lift that thing out.

Water goes in the sink and in the tank, and nowhere else. The kettle and the jug are wet after the tap, so you hold each one over the bowl to drip before you carry it. Some drops still stay on the outside, and the last step wipes them off. Water on the dispenser body, on the counter, or on the floor is a spill. You do not wipe a spill. You report it.

The order never changes: **kettle to the sink, fill it, put it back; jug to the sink, fill it, pour it into the tank, put the jug back; wipe the kettle and the jug dry.**

The arms never swap jobs. **The right gripper carries everything:** the kettle, the jug, and the cloth. It also does the pour. **The left gripper works the tap** and nothing else. It never lifts or

carries anything.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the counter. Stop it **square to the front edge of the counter**, so the front edge runs straight across the frame and both ends look the same distance away.
2. Stop it **in the middle of the sink bowl**, so the left side and the right side of the bowl are the same distance from the middle of the base.
3. Stop it **close enough** that the **right gripper** reaches the middle of the bowl floor without stretching the arm. Stop it **far enough** that no arm, wrist, or part of the base touches the counter front, the sink, or the wall cabinet while both arms work.
4. Check the **height**. With the base parked, the **right gripper** lifts the kettle just off its power base, brings it forward into the front lane, and carries it to the bowl. The wrist and the forearm must not hit the wall cabinet.
5. Check the **pour**. The **right gripper** lifts the empty fill jug off its spot, carries it to the front of the dispenser, raises it until its lip is just over the filler rim, and tips it. The jug, the wrist, and the forearm must not touch the wall cabinet or the dispenser while it does this.
6. Check the **tap**. The **left gripper** reaches the tap lever with the tap off. It lifts and drops the lever all the way without stretching and without hitting the wall cabinet or the right arm.
7. Check the **counter**. The **right gripper** reaches the kettle handle on its power base, the jug handle on its spot, the dispenser filler, and the cloth. It does all four without stretching and without knocking anything over.
8. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
9. If any line from 1 to 7 fails, push the base to a new spot by hand and start again at line 1. Do not work at a sink the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the counter hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the sink bowl. The frame shows the whole bowl, the tap and its lever, the front lane, the kettle on its power base, the dispenser with its tank and filler, the fill jug on its spot, and the cloth.
3. The camera sees the bowl from the front, so you can see where a thing stands and if it sits under the tap.
4. The camera sees the kettle's **water window**, the jug's **fill line**, and the tank's **fill line** from the front, so you can see how full each one is.
5. The camera sees the tap lever, so you can see if the tap is on or off and if it is still on cold.
6. The camera sees the top of the dispenser, so you can see the jug's lip over the filler and see the water going in during the pour.
7. The camera sees the outside of the kettle and the outside of the jug, so you can see drips on them and see if they are dry at the end.
8. Both arms are at home with the grippers open.
9. The **right arm** reaches the kettle handle on its power base, the jug handle on its spot, the dispenser filler, the middle of the bowl floor, and the cloth. It reaches all five without stretching to its limit.
10. The **left arm** reaches the tap lever all the way up and all the way down without stretching. It needs to touch nothing else.
11. The **right gripper** lifts a thing **just off** what it stands on and carries it **low and level along the front lane**. It never lifts a thing up to cabinet height. It never carries one over the power base or the dispenser. The one lift above the counter is the pour, and the jug goes up only as far as its lip needs to clear the filler rim.
12. The two arms do not hit each other. The left arm at the tap stays out of the right arm's path to the bowl.
13. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 9 to 12 are all true.

Materials checklist

1. The **sink** is one bowl set in the counter. Nobody moves it, leans on it, or stands anything on its rim.
2. The bowl is deep enough and wide enough for the kettle and the jug to stand upright on its floor, away from the walls, with the tap above their open tops.
3. The bowl drains well. The drain is clear, so water runs away.
4. There is **no wall cabinet over the sink**, so you can lower a thing straight down into the bowl.

5. There is a **wall cabinet right above the back of the counter** to the right of the sink, over the power base and the dispenser. So nothing is ever lifted high there. There is clear space between the top of the dispenser filler and the cabinet, enough for the jug to tip over the filler with its lip just over the rim and nothing touching the cabinet.
6. The **tap** is a mixer tap with one lever at the back of the bowl. Its lever is on the left. You lift the lever to start the water. You push it down to stop the water.
7. The tap is set to **cold** and stays on cold. Nobody swings the lever toward hot.
8. The **tap spout** points at the middle of the bowl. It does not turn during the episode. Nobody pushes it.
9. The **kettle** is an electric kettle. It has a handle over the top and a wide open mouth at the end of its spout. Water goes in through that mouth. Its lid stays shut. Nobody touches the lid.
10. The kettle has a **water window** down one side with a **fill line** on it. The window faces the front when the kettle stands on its power base.
11. The kettle starts the episode **below its line**. It has some water in it, but not up to the line.
12. The **power base** is the plate the kettle stands on. It is heavy and does not slip. It stands at the back of the counter, right of the sink. Nobody moves it, lifts it, or wipes under it.
13. The power base's cord runs to the back. It does not lie in the front lane.
14. The **dispenser** stands at the back of the counter, right of the power base. It is one fixed piece. Its body holds a **tank** with an **open filler** on top at the front, a **fill line** on the front of the tank that you can read from the front, its own small **tap** at the front, and a **drip tray** under that tap. Nobody moves, lifts, or tips the dispenser, and the tank never comes off the body.
15. The filler has no lid or cap. It stays open for the whole episode.
16. The tank starts the episode **below its line**, but not by more than the fill jug holds up to its own line. So one jugful always brings the tank to its line.
17. The **fill jug** is a plain jug that is not part of the dispenser. It has a **handle at the back**, a **pouring lip at the front**, an **open top**, no lid, and a **fill line** on its side that you can read from the front. It only carries water from the sink to the tank.
18. The fill jug starts the episode **empty**, standing upright on its **spot** at the right end of the counter, past the dispenser, out of the front lane, with its handle to the back and its lip to the front.
19. The jug has no lid. So you carry it low and level both ways.
20. The **front lane** is the strip of counter along the front edge, from the sink to the right end of the counter. It is empty and dry at the start. Nobody ever puts anything down in it.
21. The **cloth** is a folded, dry cloth that soaks up water. It lies flat on the counter at the right end, to the right of the jug spot, out of the front lane.
22. The counter, the bowl, the rim, the dispenser, and the floor are all dry at the start.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the sink and the counter themselves.

- **Sink bowl:** the bowl the base is parked at. Both the kettle and the jug are filled standing on its floor. **Right gripper only.**
- **Tap and lever:** the tap at the back of the bowl. **Left gripper only**, and only on the lever.
- **Front lane:** the clear strip of counter along the front edge, from the sink to the right end of the counter. Every carry goes along it. Nobody ever puts anything down in it.
- **Power base:** the kettle's plate at the back of the counter, right of the sink. Never moved. Never wiped under.
- **Kettle:** starts and ends on the power base. **Right gripper only**, on its handle.
- **Dispenser:** the fixed body with its tank, filler, own tap, and drip tray, at the back of the counter, right of the power base. Never moved. Never lifted. Never wiped. Water goes in only through the filler, only from the jug.
- **Jug spot:** the place at the right end of the counter, past the dispenser, where the fill jug stands. The jug starts and ends here.
- **Fill jug:** starts and ends upright on its spot. **Right gripper only**, on its handle.
- **Cloth:** lies folded flat at the right end of the counter, to the right of the jug spot. **Right gripper only.**
- **Wall cabinet:** the cabinet over the back of the counter. It is the reason every lift stays low. Nothing ever touches it.

Arm assignments

- **Left gripper: the tap.** It lifts the lever to start the water and drops the lever to stop the water, in Step 2 and Step 5. It touches nothing else. Not the kettle. Not the jug. Not the cloth. Not a fixture.
- **Right gripper: the working gripper.** It carries the kettle, carries the jug, holds each one still while it fills, puts the kettle back, pours the jug into the tank, puts the jug back, and wipes with the cloth.
- Nothing is passed from one gripper to the other. Only one thing moves at a time.

Vocabulary

- **Tap spout:** the part of the tap that the water comes out of.
- **Kettle mouth:** the open hole at the end of the kettle's spout. Water goes into the kettle here.
- **Tank:** the part of the dispenser that holds the water. It is fixed to the dispenser body and never comes off.

- **Filler:** the open hole on top of the tank, at the front. Water goes into the tank here, and only here.
- **Fill jug:** the separate jug that carries water from the sink to the tank. It is not part of the dispenser.
- **Lip:** the pouring edge at the front of the jug. Water comes out of the jug here.
- **Jug spot:** the place at the right end of the counter, past the dispenser, where the jug stands at the start and at the end.
- **Low carry:** you lift the thing just off what it stands on, bring it forward into the front lane, and carry it along the lane close over the counter and level. You never lift it toward the wall cabinet. You never carry it over the power base or the dispenser.
- **Front lane:** the clear strip of counter along the front edge, from the sink to the right end of the counter. Every carry goes along it. Nobody puts anything down in it.
- **Under the tap:** the thing stands upright on the bowl floor with its open top right under the tap spout. Water falls straight in. It does not fall on the outside.
- **Fill line:** the mark on the kettle's water window, the mark on the jug's side, and the mark on the front of the tank. You read all three from the front.
- **Full to the line:** the top of the water is level with the line. Both read at the same height from the front. Water above the line is too much. Water below the line is too little.
- **Steady stream:** the water comes out of the tap as one smooth stream and stays inside the open top. A stream that splashes back out, sprays off the rim, or bangs down hard is too strong.
- **Tap off:** the lever is all the way down. No water comes out of the spout. Only the last drops fall.
- **On cold:** the lever stays on the cold side for the whole episode. Nobody swings it toward hot.
- **Dripped off:** you hold the thing over the bowl for 2 seconds after the tap is off. The loose water on its outside falls in the sink, not on the counter.
- **Over the filler:** the jug is raised so its lip is just above the filler rim, and moved back so the lip is over the open filler. Water tipped out of the lip falls straight into the tank.
- **Slow pour:** the jug is tipped forward a little at a time, so the water leaves the lip as one thin, smooth stream that falls inside the filler. A pour that gushes, splashes off the filler rim, or runs down the front of the tank is too fast.
- **Levelled:** the jug is tipped back until its lip is above the water inside and nothing comes out. Water left in the jug stays in the jug.
- **Sitting right on the base:** the kettle stands flat on its power base, inside the rim of the plate, upright, with its water window and handle facing the front. It does not rock when the **right gripper** lets go of the handle.

- **Sitting right on its spot:** the jug stands flat on its spot at the right end of the counter, upright, with its handle to the back and its lip to the front, out of the front lane. It does not rock when the **right gripper** lets go of the handle.
- **Drip:** a drop or a wet streak on the outside of the kettle or the jug. You wipe these off in Step 7.
- **Spill:** water on the floor, on the dispenser body, or a pool on the counter big enough to run off the edge. You do not wipe a spill. You report it.
- **Wiped dry:** the outside looks dull from the front. No shine. No drops. No wet streaks.
- **Back and clear:** the arm is pulled back, so no part of it is over the bowl, the front lane, or the counter.

Steps

Step 1: Take the kettle to the sink

Goal: the kettle stands upright on the bowl floor, under the tap, with its water window facing the front.

1.1 Lift the kettle off its base

- With the **right gripper**, close on the kettle handle in the middle.
- Lift the kettle **just off** the power base. Its bottom leaves the plate, and no higher.
- Pull it straight forward into the **front lane**. Do not swing it. Do not lift it toward the wall cabinet.

Check: the kettle hangs upright and level in the **right gripper**. The power base has not moved. No part of the arm is near the wall cabinet. If the kettle leans, put it back down on the base, take the handle again in the middle, and lift again.

1.2 Carry it to the bowl

- Carry the kettle along the **front lane** to the sink in a **low carry**. Keep it level the whole way.
- Do not carry it over the power base or the dispenser. Do not let it touch the sink rim.
- Bring it over the middle of the bowl.

1.3 Stand it under the tap

- Lower the kettle straight down onto the bowl floor.
- Put it down upright, **under the tap**, with its **water window facing the front**. Keep its body away from the bowl walls.
- Keep the **right gripper** closed on the handle.

Check: the kettle stands flat on the bowl floor. Its mouth is right under the tap spout. You can read its window from the front. It touches no bowl wall. If it stands away from the tap, or the window faces the wrong way, lift it just off the floor and put it down again.

Expected state: the kettle stands in the bowl under the tap. The **right gripper** still holds its handle. The power base is empty. The tap is still off.

Step 2: Fill the kettle to its line

Goal: the water is level with the kettle's fill line, and the tap is off.

2.1 Hold the kettle still

- Keep the **right gripper** closed on the kettle handle. Hold the kettle still on the bowl floor.
- The **right gripper** does not lift, tip, or slide the kettle while the water runs.

2.2 Start the water

- With the **left gripper**, close on the tap lever.
- Lift the lever slowly, until the water runs as a **steady stream** into the kettle mouth.
- Do not lift it higher. Do not swing the lever toward hot.
- Keep the **left gripper** closed on the lever while the water runs.

Check: the water falls into the kettle mouth as a steady stream. Nothing splashes out of the kettle or out of the bowl. The lever is still on cold. If the water splashes, drop the lever a little with the **left gripper** until the stream is smooth.

2.3 Stop the water at the line

- Watch the **water window** from the front while the kettle fills.
- When the water reaches the **fill line**, push the lever all the way down with the **left gripper**.
- Open the **left gripper**, pull it back away from the tap, and keep it away until Step 5.

Check: the water looks level with the line from the front. The tap has stopped running. The lever is all the way down. If the water went past the line, leave it and say so when you hand the station back. Do not pour water out of the kettle. If the water is below the line, lift the lever again with the **left gripper** and add a little more.

Expected state: the kettle stands in the bowl, full to its line. The tap is off. The **right gripper** still holds the handle. The **left gripper** is away from the tap.

Step 3: Put the kettle back on its base

Goal: the kettle stands on its power base, full, upright, with its window and handle facing the front.

3.1 Let it drip over the bowl

- With the **right gripper**, lift the kettle just off the bowl floor and hold it over the bowl.
- Hold it there for **2 seconds**, so the water on its outside falls in the sink.
- Keep it level. Do not tip it. Do not swing it while it drips.

Check: the drops from the kettle fall in the bowl, not on the rim and not on the counter.

3.2 Carry it back

- Carry the kettle along the **front lane** back to the power base in a **low carry**. Keep it level the whole way.
- Do not carry it over the dispenser. Do not lift it toward the wall cabinet.
- Bring it in front of the power base. Then move it straight back, over the plate.

3.3 Put it down on the plate

- Lower the kettle straight down onto the power base.
- Put it down inside the rim of the plate, upright, with its **water window and handle facing the front**.
- Open the **right gripper** and pull it back away from the kettle.

Check: the kettle is **sitting right on the base**. It is flat inside the rim, upright, with its window and handle to the front, and it does not rock when the **right gripper** lets go. If it sits on the rim or it rocks, take the handle again with the **right gripper**, lift it just off the plate, and put it down again.

Expected state: the kettle stands full on its power base. The bowl is empty. The tap is off. Both grippers are back and clear.

Step 4: Take the fill jug to the sink

Goal: the jug stands upright on the bowl floor, under the tap, with its fill line facing the front.

4.1 Lift the jug off its spot

- With the **right gripper**, close on the jug handle at the back of the jug.
- Lift the jug **just off** the counter. Its bottom leaves the counter, and no higher.
- Pull it straight forward into the **front lane**. Do not lift it toward the wall cabinet. Do not knock the dispenser or the cloth.

Check: the jug hangs upright and level in the **right gripper**. The dispenser has not moved. The cloth has not moved. If the jug leans, put it back down on its spot, take the handle again, and lift again.

4.2 Carry it to the bowl

- Carry the jug along the **front lane** to the sink in a **low carry**. Keep it level the whole way.
- Do not carry it over the dispenser, the power base, or the kettle. Do not let it touch the sink rim.
- Bring it over the middle of the bowl.

4.3 Stand it under the tap

- Lower the jug straight down onto the bowl floor.
- Put it down upright, **under the tap**, with its **fill line facing the front**. Keep its body away from the bowl walls.
- Keep the **right gripper** closed on the handle.

Check: the jug stands flat on the bowl floor. Its open top is right under the tap spout. You can read its line from the front. It touches no bowl wall. If it stands away from the tap, or the line faces the wrong way, lift it just off the floor and put it down again.

Expected state: the jug stands in the bowl under the tap. The **right gripper** still holds its handle. The jug spot is empty. The tap is still off.

Step 5: Fill the jug to its line

Goal: the water is level with the jug's fill line, and the tap is off.

5.1 Hold the jug still

- Keep the **right gripper** closed on the jug handle. Hold the jug still on the bowl floor.
- The **right gripper** does not lift, tip, or slide the jug while the water runs.

5.2 Start the water

- With the **left gripper**, close on the tap lever.
- Lift the lever slowly, until the water runs as a **steady stream** into the jug's open top.
- Do not lift it higher. Do not swing the lever toward hot.
- Keep the **left gripper** closed on the lever while the water runs.

Check: the water falls inside the jug's top as a steady stream. Nothing splashes out of the jug or out of the bowl. The lever is still on cold. If the water splashes, drop the lever a little with the **left gripper** until the stream is smooth.

5.3 Stop the water at the line

- Watch the **fill line** from the front while the jug fills.
- When the water reaches the line, push the lever all the way down with the **left gripper**.
- Open the **left gripper**, pull it back away from the tap, and keep it away for the rest of the episode.

Check: the water looks level with the line from the front. The tap has stopped running. The lever is all the way down. If the water went past the line, leave it and say so when you hand the station back. Do not pour water out of the jug into the sink. If the water is below the line, lift the lever again with the **left gripper** and add a little more.

Expected state: the jug stands in the bowl, full to its line. The tap is off. The **right gripper** still holds the handle. The **left gripper** is away from the tap.

Step 6: Pour the jug into the dispenser

Goal: the tank reads full to its line, the jug is levelled and back on its spot, and no water is on the dispenser or the counter.

6.1 Let it drip over the bowl

- With the **right gripper**, lift the jug just off the bowl floor and hold it over the bowl.
- Hold it there for **2 seconds**, so the water on its outside falls in the sink.
- Keep it level. Do not tip it. Do not swing it while it drips.

Check: the drops from the jug fall in the bowl. The water inside is not slopping out of the open top.

6.2 Carry it to the dispenser

- Carry the jug along the **front lane** to the dispenser in a **low carry**. Keep it level the whole way.
- Do not carry it over the kettle or its power base. Do not lift it toward the wall cabinet.
- Stop it in the **front lane**, right in front of the dispenser, still level and still low.

6.3 Raise it over the filler

- Lift the jug straight up, in front of the dispenser, until its **lip is just above the filler rim**, and no higher.
- Move it straight back until the lip is **over the filler**. Keep the jug level while it moves.
- Do not touch the tank, the filler rim, or the dispenser body with the jug. Do not let the jug, the wrist, or the forearm go near the wall cabinet.

Check: the lip is over the open filler and just above its rim. The jug is still level, and nothing has come out of it yet. Nothing touches the dispenser or the cabinet. If the lip is beside the filler instead of over it, move the jug sideways until it is over. If the jug is higher than it needs to be, lower it until the lip is just above the rim.

6.4 Pour slowly to the tank's line

- Tip the jug forward by turning the wrist, a little at a time, until the water leaves the lip as a **slow pour** and falls inside the filler.
- Keep the lip over the filler the whole time. Do not move the jug along while it pours.
- Watch the **tank's fill line** from the front while the tank fills.
- When the water in the tank reaches the line, tip the jug back until it is **levelled** and nothing comes out.
- Do not tip the jug further to empty it. Any water left in the jug stays in the jug.

Check: the water in the tank looks level with the line from the front. The jug is levelled. No water is on the front of the tank, on the dispenser body, on the drip tray, or on the counter. If the

water went past the line, leave it and say so when you hand the station back. If the water is below the line and the jug still has water in it, tip the jug forward again and add a little more.

6.5 Put the jug back on its spot

- Move the jug straight forward, still level, until it is clear of the dispenser. Then lower it straight down into the **front lane**, just above the counter.
- Carry it along the **front lane** to the right end of the counter in a **low carry**. Keep it level the whole way.
- Put it down upright on its **spot**, past the dispenser, out of the lane, with its **handle to the back and its lip to the front**.
- Open the **right gripper** and pull it back away from the jug.

Check: the jug is **sitting right on its spot**. It stands flat, upright, handle to the back and lip to the front, out of the front lane, and it does not rock when the **right gripper** lets go. If it stands in the lane or it rocks, take the handle again with the **right gripper**, lift it just off the counter, and put it down again.

Expected state: the kettle stands full on its power base. The tank reads full to its line. The jug stands on its spot. The bowl is empty. The tap is off. The only water left outside is drips on the outside of the kettle and the jug.

Step 7: Wipe the kettle and the jug dry

Goal: the outside of the kettle and the outside of the jug are dry, and the cloth is back where it started.

Wipe only these two things. You do not wipe the counter, the front lane, the sink, the sink rim, or the dispenser in this step.

7.1 Pick up the cloth

- With the **right gripper**, close on the cloth at its near edge. Lift it just off the counter.
- Bring it along the **front lane** to the kettle.

7.2 Wipe the kettle

- Wipe the outside of the kettle with the cloth. Wipe the sides, the handle, and the spout.
- Wipe with the cloth flat. Do not bunch it into a ball.
- Wipe gently. Do not push the kettle, and do not lift it off its power base.

Check: the outside of the kettle is **wiped dry**. No shine. No drops. No wet streaks. The kettle has not moved on its base. If a wet spot is left, wipe that spot again.

7.3 Wipe the jug

- Carry the cloth along the **front lane** to the jug on its spot.
- Wipe the outside of the jug with the cloth. Wipe the sides, the handle, and the lip.

- Wipe gently. Do not push the jug, and do not lift it off its spot.

Check: the outside of the jug is **wiped dry**. No shine. No drops. No wet streaks. The jug has not moved on its spot. If a wet spot is left, wipe that spot again.

7.4 Put the cloth back

- Lay the cloth down flat where it started, to the right of the jug, out of the front lane.
- Open the **right gripper** and pull it back.

Expected state: the kettle and the jug are dry outside. The cloth lies flat where it started.

Step 8: End the episode

Goal: the kettle and the tank are full, the jug and the cloth are back on their spots, the tap is off, and both arms are safely home.

- Confirm the kettle stands full on its power base and the tank reads full to its line.
- Confirm the tap lever is all the way down, no water runs, and nothing is spilled on the dispenser, the counter, or the floor.
- Confirm the base has not moved.
- Return both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Pour water out of the kettle into the sink, until the water is **below its line**. Dry the outside of the kettle. Stand it back on its power base, window and handle to the front.
2. Draw water off the tank through the dispenser's own tap into the fill jug, until the tank is **below its line** by less than one jugful. Pour that water into the sink.
3. Empty the fill jug into the sink. Dry it inside and out. Stand it back on its spot at the right end of the counter, handle to the back and lip to the front.
4. Empty the dispenser's drip tray and dry it. Dry the top of the tank around the filler and the front of the tank.
5. Dry the bowl floor, the sink rim, the counter front edge, and the whole front lane.
6. Hang the used cloth up to dry. Lay a dry folded cloth to the right of the jug spot, out of the front lane.
7. Check the tap: the lever goes all the way down and stops the water fully. Leave the lever on cold.
8. Check the tap spout still points at the middle of the bowl and has not been turned.
9. Check the bowl drains well and no water stands on its floor.
10. Change the kettle if its water window has gone cloudy, if you cannot read its line from the front, or if it rocks on its power base.

11. Change the jug if you cannot read its line from the front, if its lip is chipped so it dribbles, or if it will not stand flat on the counter.
12. Check the dispenser: the tank line can be read from the front, the filler is open and clear, its own tap does not drip, and no water sits on the body.
13. Check the power base and the dispenser still stand at the back of the counter, square and not slipping, with the cord tucked to the back and out of the front lane.
14. Wipe up any spill on the floor and dry it before the next episode.
15. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the **start timestamp**, the **violation name**, and the **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the sink moves in the frame, the counter edge changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.
- **SOP rule broken:** Steps 1 to 8. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Lifted high or hit the wall cabinet

- **Visible cue:** the **right gripper** lifts the kettle or the jug well above the counter instead of just off what it stands on, carries it at cabinet height, or raises the jug for the pour far higher than its lip needs to clear the filler rim. Or a wrist, a forearm, or the jug knocks, scrapes, or rests on the wall cabinet.
- **SOP rule broken:** Steps 1.1, 1.2, 3.2, 4.1, 4.2, 6.2, 6.3, and 6.5. Lift just off, come forward into the front lane, and carry low and level. For the pour, go up only until the lip clears the rim.

- **Coaching note:** up a little, forward, then along. The pour goes up just enough and no more. The cabinet is right there.

Violation: Carried outside the front lane

- **Visible cue:** the **right gripper** carries the kettle, the jug, or the cloth over the power base, over the dispenser, or across the back of the counter instead of along the front lane. Or it puts a thing down in the lane in the middle of a carry.
- **SOP rule broken:** Steps 1.2, 3.2, 4.2, 6.2, 6.5, 7.1, and 7.3. Every carry goes along the front lane, and nothing is ever put down in it.
- **Coaching note:** the lane is the road. Nothing crosses the back of the counter, and nothing parks in the lane.

Violation: Leaned on or moved a fixture

- **Visible cue:** a gripper, wrist, or forearm rests on the sink, the tap, the power base, or the dispenser. Or the jug rests on the filler rim or the tank during the pour. Or the power base or the dispenser slides or rocks. Or the tap spout is pushed or turned.
- **SOP rule broken:** Steps 1 to 7. Nobody leans on, pushes, drags, or lifts the sink, the tap, the power base, or the dispenser. The jug pours from above the filler and never rests on it.
- **Coaching note:** the arm holds itself up, and so does the jug. If something moves when you touch it, you touched it too hard.

Violation: Wrong gripper used

- **Visible cue:** the **left gripper** lifts, carries, pours, puts back, or holds the kettle, the jug, or the cloth. Or the **right gripper** touches the tap lever. Or one arm reaches across the body to do the other arm's job.
- **SOP rule broken:** Steps 1 to 7. The right gripper carries, pours, and puts back everything. The left gripper works only the tap lever.
- **Coaching note:** right hand carries and pours, left hand taps. Nothing swaps, not even once.

Violation: Held by the wrong part

- **Visible cue:** the **right gripper** closes on the kettle's body, spout, or lid instead of its handle. Or on the jug's rim, body, or lip instead of its handle. Or a thing is dragged along the counter instead of lifted.
- **SOP rule broken:** Steps 1.1, 3.3, 4.1, and 6.5. Hold the kettle by its handle and the jug by its handle, and lift, do not drag.
- **Coaching note:** handles only, both of them, both ways. A dragged kettle scratches and slops.

Violation: Carried leaning or slopping

- **Visible cue:** the kettle or the jug hangs at an angle during a carry, or it swings, or water slops out of the jug's open top or out of the kettle mouth while it moves. Or the jug tips and lets water out anywhere other than over the filler.
- **SOP rule broken:** Steps 1.2, 3.2, 4.2, 6.2, 6.3, and 6.5. Carry it level the whole way. The jug tips only over the filler.
- **Coaching note:** level all the way there and all the way back. The jug has no lid, so a lean is a spill.

Violation: Filled in the air

- **Visible cue:** the **right gripper** holds the kettle or the jug up under the tap while the water runs instead of standing it on the bowl floor. Or it lifts one off the floor before the tap is off.
- **SOP rule broken:** Steps 1.3, 2.1, 2.3, 4.3, 5.1, and 5.3. The thing stands on the bowl floor before the water starts, and you lift it only after the water is off.
- **Coaching note:** stand it down, then turn the water on. The tap never runs into anything hanging in the air.

Violation: Stood wrong in the bowl

- **Visible cue:** the kettle or the jug stands to one side of the tap, so water lands on its outside. Or it leans on a bowl wall. Or it stands over the drain. Or its window or fill line faces away from the front.
- **SOP rule broken:** Steps 1.3 and 4.3. Stand it upright under the tap, away from the bowl walls, with its window or line facing the front.
- **Coaching note:** open top under the tap, mark to the front. If you cannot see the line, you cannot fill to it.

Violation: Water run too hard

- **Visible cue:** the **left gripper** lifts the lever so high that the water gushes, sprays off the rim, splashes back out, or throws water out of the bowl onto the counter.
- **SOP rule broken:** Steps 2.2 and 5.2. Lift the lever slowly until the water runs as a steady stream, and no higher.
- **Coaching note:** slowly up until it is one smooth stream. A gush is not faster, and it wets everything.

Violation: Lever moved off cold

- **Visible cue:** the **left gripper** swings the tap lever toward the hot side at any time. Or the lever is off cold at the end of the episode.
- **SOP rule broken:** Steps 2.2 and 5.2. The lever stays on cold for the whole episode.
- **Coaching note:** the lever only goes up and down. Sideways is not part of this task.

Violation: Water left running

- **Visible cue:** the tap runs with nothing under it. Or the **left gripper** leaves the lever up while the arm goes somewhere else. Or the water is still running when a thing is lifted out of the bowl.
- **SOP rule broken:** Steps 2.3 and 5.3. Push the lever all the way down before you lift anything, and never leave the water running.
- **Coaching note:** water off first, hands second. The tap is never left on by itself.

Violation: Poured outside the filler or too fast

- **Visible cue:** the **right gripper** tips the jug before its lip is over the filler, or moves the jug along while it pours, so water lands on the top of the tank, runs down the front of the tank, or falls on the drip tray or the counter. Or the jug is tipped so far that the water gushes and splashes off the filler rim.
- **SOP rule broken:** Steps 6.3 and 6.4. Get the lip over the filler first, then tip a little at a time, and keep the lip over the filler the whole time.
- **Coaching note:** over first, then tip. A thin stream straight into the hole. If you see water on the tank, you poured too soon or too fast.

Violation: Filled past the line or below it

- **Visible cue:** when the **left gripper** drops the lever, the water in the kettle's window or in the jug is clearly above the fill line or clearly below it. Or when the **right gripper** levels the jug, the water in the tank is clearly above the tank's line or clearly below it. Or the jug is emptied into the tank with no look at the tank's line.
- **SOP rule broken:** Steps 2.3, 5.3, and 6.4. Stop the water, or level the jug, when the water is level with the fill line.
- **Coaching note:** watch the line, not the clock. Stop at the line, not after it. The jug does not have to be emptied.

Violation: Fill not watched from the front

- **Visible cue:** the **left gripper** drops the lever, or the **right gripper** levels the jug, without any look at the window or the line. Or the mark is turned away or hidden by the arm, so nobody can read the fill from the front at any time.
- **SOP rule broken:** Steps 2.3, 5.3, and 6.4. Watch the window, the jug's line, or the tank's line from the front while it fills.
- **Coaching note:** make sure you can see the line before the water moves. A fill you cannot see is a guess.

Violation: Carried wet without dripping off

- **Visible cue:** the **right gripper** lifts the kettle or the jug straight out of the bowl and carries it away with no hold over the bowl. Or a line of drops is left across the counter behind a carry.

- **SOP rule broken:** Steps 3.1 and 6.1. Hold it over the bowl for 2 seconds, so the water on its outside falls in the sink.
- **Coaching note:** two seconds over the bowl. Every drop you leave there is one you do not wipe later.

Violation: Kettle not sitting right on its base

- **Visible cue:** the kettle ends up on the rim of the power base, part off the plate, rocking when the **right gripper** lets go, with its window or handle turned away from the front, or left on the counter instead of on the base.
- **SOP rule broken:** Step 3.3. Put the kettle down flat inside the rim of the plate, upright, window and handle to the front, so it does not rock.
- **Coaching note:** straight down inside the rim, then watch it settle before you let go.

Violation: Jug not sitting right on its spot

- **Visible cue:** the jug ends up leaning, rocking when the **right gripper** lets go, standing in the front lane, standing in front of the dispenser, standing on the cloth, with its lip turned away from the front, or left in the bowl instead of on its spot.
- **SOP rule broken:** Step 6.5. Put the jug down flat on its spot at the right end of the counter, upright, handle to the back and lip to the front, out of the front lane, so it does not rock.
- **Coaching note:** all the way back to its spot, lip to the front. A jug left in the lane is in the way of the cloth.

Violation: Done in the wrong order

- **Visible cue:** the jug goes to the sink before the kettle is full and back on its base. Or the kettle and the jug are both in the bowl at once. Or the jug goes to the dispenser before it has been filled at the sink. Or the cloth comes out before the kettle is on its base and the jug is on its spot.
- **SOP rule broken:** Steps 1 to 7. Kettle first, then the jug to the sink, then the jug to the tank, then the wipe. One thing at a time.
- **Coaching note:** one thing from its place to the sink and back before the next one starts. The wipe is always last.

Violation: Kettle or jug left wet

- **Visible cue:** at the end of Step 7 the outside of the kettle or the outside of the jug still shows shine, drops, or wet streaks. Or the wipe misses one of the two things.
- **SOP rule broken:** Steps 7.2 and 7.3. Wipe the outside of the kettle and the outside of the jug until each one is dry.
- **Coaching note:** both of them, every time. Look along the side for shine before you stop.

Violation: Wiped the wrong thing

- **Visible cue:** the **right gripper** wipes the counter, the front lane, the sink, the sink rim, or the dispenser in Step 7.
- **SOP rule broken:** Step 7. Only the outside of the kettle and the outside of the jug are wiped.
- **Coaching note:** two things get wiped, and only two. The counter and the dispenser are not part of this step.

Violation: Kettle or jug moved during the wipe

- **Visible cue:** the **right gripper** lifts, pushes, or slides the kettle, the jug, the power base, or the dispenser during the wipe. Or the cloth drags one of them out of place.
- **SOP rule broken:** Steps 7.2 and 7.3. Wipe gently, and do not move the kettle or the jug to wipe it.
- **Coaching note:** hold the cloth, not the kettle. Nothing on the counter moves in Step 7.

Violation: Cloth used or put back wrong

- **Visible cue:** the **right gripper** drags the cloth across the counter instead of lifting it, wipes with it bunched in a ball, drops it in the sink or on the floor, leaves it on the front lane, or lays it down somewhere other than to the right of the jug spot.
- **SOP rule broken:** Steps 7.1 and 7.4. Lift the cloth by its near edge, wipe with it flat, and lay it back flat where it started, to the right of the jug and out of the front lane.
- **Coaching note:** lift it, wipe flat, lay it back in the same spot. The cloth ends where it began.

Violation: Water spilled outside the sink

- **Visible cue:** the kettle or the jug is knocked over, water pools on the counter big enough to run off the edge, water runs down the dispenser body onto the counter, or water goes on the floor.
- **SOP rule broken:** Steps 1 to 7. Water goes in the sink and in the tank, and nowhere else.
- **Coaching note:** stop, report it, and reset. You do not wipe a spill away in the middle of an episode.

Violation: Something dropped or knocked over

- **Visible cue:** the kettle, the jug, or the cloth slips out of the **right gripper**, falls in the bowl or on the counter, or is knocked over by an arm going past.
- **SOP rule broken:** Steps 1 to 7. One thing moves at a time, and each one is put down under control.
- **Coaching note:** close on the handle all the way before you lift, and put it down before you let go.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails, and the episode carries on with no retry. For example, the kettle stands away from the tap, the fill goes past the line, the jug's lip is beside the filler, the kettle rocks on its base, or a wet spot is left.
- **SOP rule broken:** Steps 1 to 7. Do the retry written under each failed check before you move on.
- **Coaching note:** a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

- **Visible cue:** the episode ends with the kettle off its base, the jug off its spot, the kettle or the tank below its line, or the kettle or the jug still wet. Or the tap is up or off cold. Or the kettle or the jug is still in the bowl. Or the cloth is out of place. Or an arm is not home, or a gripper is not fully open.
- **SOP rule broken:** Step 8. Confirm the kettle, the tank, the tap, and the base. Then return both arms home with the grippers open and stop recording.
- **Coaching note:** confirm first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see the water window, the jug's fill line, the tank's fill line, or the tap lever**, so nobody can tell if a thing was filled to its line or if the tap was shut off.
- **Hardware fault on an arm:** gripper failure, drift, a crash caused by the controller, or a motor error.
- **Base fault:** a lock that will not hold, or a base that rolls, slides, or turns with the lock on.
- **Water supply fault:** no water at the tap, pressure that jumps or dies during a fill, or water that comes out warm with the lever on cold.
- **Faulty tap:** a lever that will not stop the water fully, that jumps from a trickle to a gush with no steady stream in between, that swings off cold on its own, or a spout that turns off the bowl by itself.
- **Blocked drain:** a bowl that will not drain, so water stands on its floor and nothing can be stood down flat.
- **Faulty kettle:** one with a cloudy or unreadable water window, a worn off line, a leak, or one that rocks on a correctly placed power base.
- **Faulty power base:** one that slides on the counter under a correct set down, or that will not hold the kettle flat.

- **Faulty jug:** one with a worn off fill line, a handle that comes loose in a correct grip, a chipped lip that dribbles down its own side during a correct slow pour, or one that will not stand flat on the counter.
- **Faulty dispenser:** a tank that leaks, a filler that is blocked or too narrow for a correct slow pour, a tank line that cannot be read from the front, its own tap dripping, or a body that slides on the counter under a correct pour.
- **Faulty cloth:** one that leaves fluff, one that will not soak up water, or one that arrives at the station already wet.
- **A place turns out to be too far for its arm** with the base parked correctly, so the kettle handle, the jug handle, the filler, the tap lever, the middle of the bowl floor, or the cloth cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Lift the kettle off its power base
2. Carry the kettle along the front lane to the sink
3. Stand the kettle under the tap
4. Lift the tap lever to start the water
5. Push the tap lever down to stop the water
6. Hold the kettle over the bowl to drip
7. Carry the kettle back along the front lane
8. Put the kettle down on its power base
9. Lift the fill jug off its spot
10. Carry the jug along the front lane to the sink
11. Stand the jug under the tap
12. Hold the jug over the bowl to drip
13. Carry the jug along the front lane to the dispenser
14. Raise the jug over the filler
15. Pour the jug into the tank to its line
16. Put the jug down on its spot
17. Pick up the cloth
18. Wipe the outside of the kettle dry
19. Wipe the outside of the jug dry
20. Lay the cloth back at the end of the counter
21. Send both arms home and end the episode